Satyam Das

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linkedin | github
Work Rights: Student Visa 500 | 48 hrs/fortnight

CAREER PROFILE

- Generative AI Developer and Computer Science graduate with a strong foundation in architecting production-grade, scalable AI systems and multi-modal industrial applications.
- Proven track record in engineering highly responsive applications, successfully reducing system latency by 90% and automating over 90% of manual business workflows.
- Published researcher with expertise in multi-agent LLMs, IoT-compatible AI, and voice-based security systems, with papers presented at international conferences like ICTIS Bangkok.
- Highly skilled in full-stack development (React, Node.js), cloud architecture (AWS Certified), and advanced machine learning frameworks including PyTorch, TensorFlow, and LangChain.
- Experienced in high-accuracy vulnerability detection systems, achieving significant benchmark improvements using hybrid GNN and RAG pipelines.

EDUCATION

Vellore Institute of Technology (VIT) — Vellore, India
B.Tech in Computer Science and Engineering — *Graduated: 2025*

- CGPA: 8.17 / 10.0

Bhavans G B Kanoria Vidya Mandir — Kolkata, India
Class 12 (CBSE) | Class 10 (CBSE) — *July 2021 | May 2019*

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript, TypeScript, HTML/CSS
AI/ML & Cloud: TensorFlow, PyTorch, LangChain, Hugging Face, OpenAI GPT, AWS (Lambda, S3, EC2, Amplify, Bedrock, Step Functions), Docker, Git
Web Development: React.js, Next.js, Node.js, Express.js, MongoDB, Tailwind CSS
Data Science: Pandas, NumPy, Financial Modelling, Vector Databases

EXPERIENCE

Workmates Core2Cloud Solutions Ltd. — Kolkata, India
Generative AI Developer Intern — *May 2025 – Dec 2025*

- Architected production-grade, scalable AI systems, automating over 90% of manual business workflows.
- Engineered highly responsive AI applications, slashing system latency by 90% for real-time users.
- Developed advanced multi-modal AI (voice/video) applications for industrial use cases.

RESEARCH-DRIVEN WORK

- **Edge-Compatible GPT-2 for IoT Devices:** Converted GPT-2 to TensorFlow Lite, achieving generative output on low-resource devices while prioritizing privacy and latency.
- **LLM-based Smart Assistant for IoT Control (Published, ISBN: 978-93-94638-96-9, GICS'24):** Leveraged OpenAI & LLAMA APIs to automate real-world tasks via natural language, enabling secure function-calling with contextual memory.
- **Financial Trend Analyzer using Multi-Agent LLMs (Published, ICTIS 2025, springer):** Built LangChain-powered pipeline to analyze financial news via autonomous LLM agents, visualizing real-time market sentiments.
- **Voice-Based Key Generation using Fuzzy Extractors (Published, De Gruyter Brill, degruyterbrill):** Built MFCC-based voice-AES key generation system (0.0185 ms latency). Reduced false acceptance by 34% using averaged thresholding.

RELEVANT PROJECTS

The Code Doctor: Advanced Vulnerability Detection

LLMs, GNNs, RAG, FAISS

- Developed a State-of-the-Art vulnerability detection system, achieving 74% accuracy on the CodeXGLUE benchmark, representing a 32% relative improvement over GNN baselines.
- Engineered a hybrid pipeline using Graph Neural Networks (GCN, GAT) and GPT-4o-mini, implementing RAG with a FAISS vector index of 20,000+ code snippets for semantic retrieval.
- Demonstrated mastery in representation learning by projecting code into low-dimensional embeddings (all-MiniLM-L6-v2) to identify disentangled patterns of logic and security defects.

The Neural Wind Tunnel: PINN-Based Fluid Dynamics

PINNs, WebGL, React Three Fiber

- Built a ”Neural Physics Engine” using Physics-Informed Neural Networks (PINNs) to solve Navier-Stokes equations in <50ms, achieving a 1000x speedup over traditional numerical CFD methods.
- Implemented Inverse Design capabilities, utilizing gradient descent on slider inputs to allow the AI to automatically optimize geometry for minimized turbulence.
- Developed a holographic 3D visualization platform using React Three Fiber to project real-time 2D neural field simulations onto 3D car models for interactive design analysis.

Conversational AI-Powered CRM Platform (LeadSpark)

React, Node.js, LiveKit, LangChain

- Developed a full-stack unified lead management system integrating a real-time, bidirectional AI voice agent using LiveKit and SIP for PSTN connectivity.
- Leveraged GPT-4, LangChain, and FAISS vector search to enable context-aware natural voice interactions with dynamic RAG-based knowledge retrieval.
- Built a secure React command center with JWT authentication and real-time analytics (Recharts) to visualize agent performance and conversion rates.

Dynamic Multi-Factor Portfolio Optimization Engine

Python, Pandas, NumPy

- Engineered a two-stage computational model analyzing 750+ equities using six proprietary factors, including Momentum, Quality, and a custom ’Special Situation’ heuristic.
- Built an optimization engine to backtest factor-weight permutations, identifying historically optimal investment models for any given period.
- Developed an ’Integrated Reasoning Engine’ to algorithmically generate human-readable, auditable justifications for every asset selection, solving the quantitative ”black box” problem.

Serverless Intelligent Rostering Agent

AWS Lambda, Step Functions, Bedrock

- Architected an asynchronous, event-driven rostering system using AWS Step Functions to orchestrate Lambda ”workers” for daily task identification.
- Implemented an intelligent ”best-match” algorithm scoring staff based on proximity, care continuity, and language preferences.
- Integrated Amazon Bedrock to enhance decision-making logic and provide detailed justifications for generated rosters.

Modular Admin Dashboard UI

React, Modern JavaScript, Vercel CI/CD

- Developed a fully responsive React application delivering a modular admin dashboard UI complete with dynamic routing, data-driven widgets, and interactive chart components.
- Showcased 74 iterative commits for a clean, well-structured codebase, leveraging zero-config Vercel CI/CD deployment.

PROFESSIONAL DEVELOPMENT

- **AWS Certified Solutions Architect Associate** (November 2023)
- **AWS Certified Cloud Practitioner** (September 2023)
- **Paper Published** in Global Innovations in Computer Science (GICS), 2024
- **Research Paper Presented and Published** in ICTIS Bangkok, 2025
- **Paper Published** in De Gruyter Brill, 2026